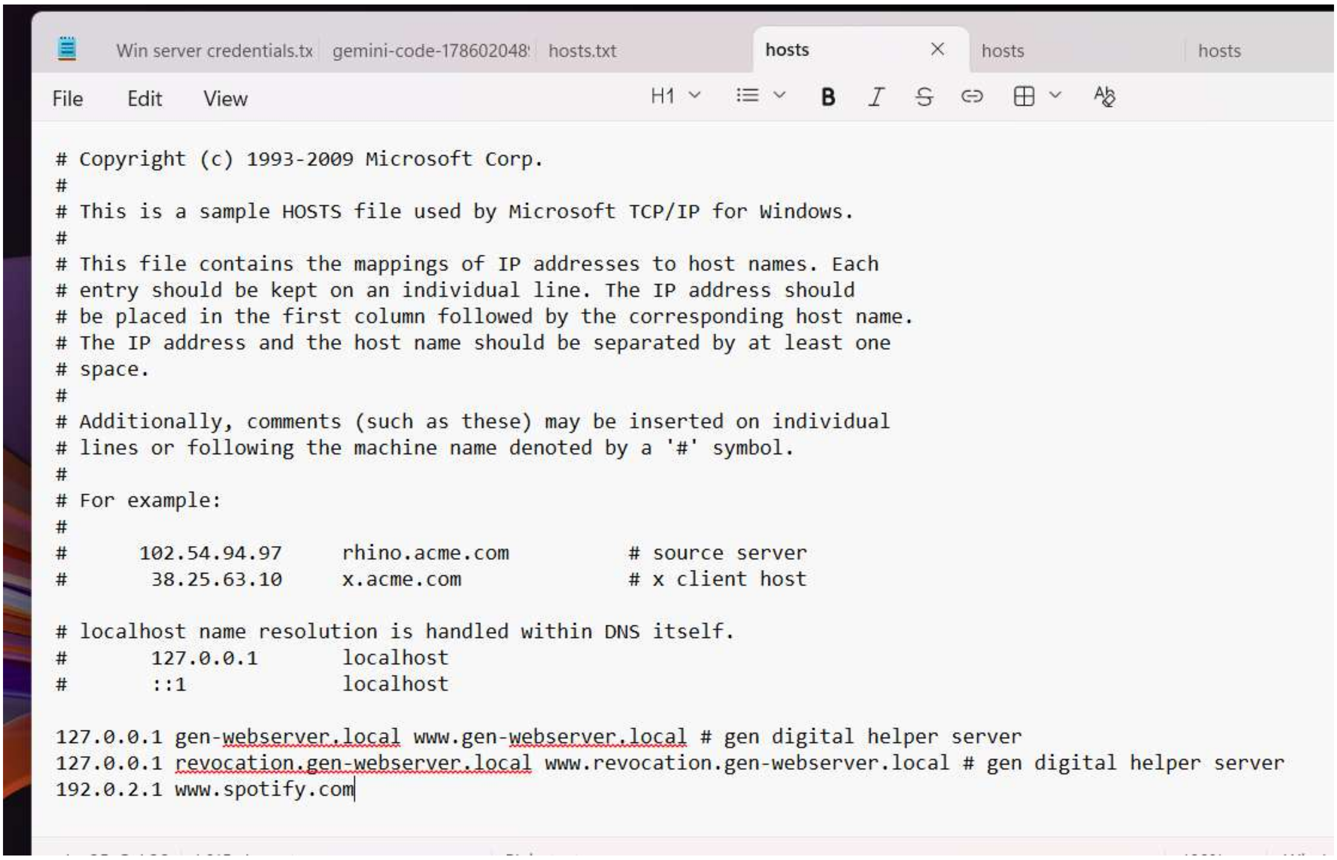
Day 5 : Troubleshooting network practicals

Task 1. Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

DNS Resolution Failure via `hosts` File

When you map a domain to an incorrect or unreachable IP address in your local `hosts` file, your operating system prioritizes this file over external DNS queries.

- **Configuration:** Add the line `192.0.2.1 www.spotify.com` to your OS `hosts` file (e.g., `C:\Windows\System32\drivers\etc\hosts` or `/etc/hosts`).




```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#      102.54.94.97     rhino.acme.com          # source server
#       38.25.63.10     x.acme.com              # x client host

# localhost name resolution is handled within DNS itself.
#       127.0.0.1       localhost
#       ::1             localhost

127.0.0.1 gen-webserver.local www.gen-webserver.local # gen digital helper server
127.0.0.1 revocation.gen-webserver.local www.revocation.gen-webserver.local # gen digital helper server
192.0.2.1 www.spotify.com
```

- **Command:** `ping www.spotify.com`
- **Error Message:**

- **Windows:** `Request timed out.` (if the fake IP doesn't respond) or `Ping request could not find host [www.spotify.com](https://www.spotify.com). Please check the name and try again.`

```
C:\Users\srikantm05>ping www.spotify.com

Pinging www.spotify.com [192.0.2.1] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.0.2.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\srikantm05>
```

Task 2.Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

```
Router(dhcp-config)#default-router 10.0.0.99
Router(dhcp-config)#do wr
Router(dhcp-config)#
```

```
                                          0.0.0.0

C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..: wr
   Physical Address................: 0002.17B9.81E5
   Link-local IPv6 Address.........: FE80::202:17FF:FEB9:81E5
   IPv6 Address....................: ::
   IPv4 Address....................: 10.0.0.4
   Subnet Mask.....................: 255.0.0.0
   Default Gateway.................: ::
                                     10.0.0.99
   DHCP Servers....................: 10.0.0.1
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-5A-7D-B6-BA-00-02-17-B9-81-E5
   DNS Servers.....................: ::
                                     30.0.0.3
```

IP config in PC0

- From IP config in PC0, Default gateway is incorrect compared to router IP address.
- pings to WAN/external IPs (e.g., `30.0.0.3`) fail with `Request timed out` because traffic is directed to non-existent gateway `10.0.0.99`.

Task 3.Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

- unable to access websites by name,
- but can by IP addresses

Run nslookup on PC0: nslookup www.wikipedia.org

- **Analyze Output:**
  - **Error:** DNS request timed out. PC0 cannot reach the DNS server IP (30.0.0.3) over port 53.
  - **Error:** *** UnKnown can't find www.wikipedia.org: Server failed - The DNS server is reachable, but the **A Record** is missing or the DNS service is turned off on Server0.
- **Solution:**
  - **If DNS Server IP is wrong on PC:** Verify ipconfig /all shows DNS Server as 30.0.0.3. If wrong, update dns-server 30.0.0.3 under Router0's DHCP pool.
  - **If DNS Service is Down/Misconfigured:** Open **Server0 - Services - DNS**, turn service **ON**, and verify www.wikipedia.org points to 30.0.0.3.

Task 4.A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

Possible Causes

- **Physical/Link Layer Failure:** Cable disconnected, switch port down, or wrong VLAN assigned.
- **Server Unavailability:** DHCP service stopped, or the router/server is offline.
- **Scope Exhaustion:** The DHCP pool ran out of available IP addresses to lease.
- **DHCP Relay / Helper Address Missing:** DHCP requests cannot cross router boundaries to a remote DHCP server.

Step-by-Step Resolution Process

1. **Check Layer 1/2 Connectivity:** Verify green link lights on the switch and PC interface. Ensure the interface is not administratively down (`no shutdown`).
2. **Verify Server Status & Scope:**
   - On **Router0**, run `show ip dhcp binding` to check active leases.
   - Check if the scope range is full. If exhausted, increase the network subnet mask or clear bindings using `clear ip dhcp binding *`.
3. **Check DHCP Service:** Ensure `service dhcp` is enabled on the Cisco router.
4. **Force Request:** On client PC Command Prompt, run:
   - ipconfig /release
   - ipconfig /renew

Task 5.Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Protocol	**Issue / Problem**	**Symptoms**	**Possible Causes**	**Recommended Solution**
DNS	Incorrect DNS Server IP	Browsing by domain fails; browsing by raw IP works.	Wrong DNS IP assigned via static setup or DHCP pool.	Update DNS server address in DHCP scope settings or local adapter properties.
DNS	Missing Resource Record	nslookup returns Non-existent domain or NXDOMAIN.	A Record for the domain was not created on the target DNS server.	Add the corresponding A Record mapping the domain to its IP address on the DNS server.
DNS	Stale Local Cache	Domain resolves to an outdated/old IP address.	Local OS DNS cache holds old records after server IP changed.	Flush local DNS cache using ipconfig /flushdns (Windows) or dscacheutil -flushcache (macOS).
DHCP	Scope Exhaustion	New devices assign themselves APIPA (169.254.x.x) addresses.	All IP addresses in the DHCP pool range are actively leased out.	Expand the subnet mask/IP pool range, or reduce lease time duration.
DHCP	Misconfigured Gateway	LAN devices can talk locally, but cannot access	default-router address under DHCP pool is set incorrectly or missing.	Update default-router <Correct_IP> in the router's DHCP pool configuration.

		external networks/Internet.		
DHCP	Rogue DHCP Server	Devices receive incorrect IP subnets or wrong gateway/DNS information randomly.	An unauthorized router/DHCP server is connected to the same switch.	Enable DHCP Snooping on switch ports to block untrusted DHCP responses.